PROBLEM K8

Pulsating Three-Dimensional Potential Flow

Original statement · K8.pdf

Potential flow | Fluid surface | Path lines | Unsteady

OFFICIAL PROBLEM STATEMENT

The velocity potential of a three-dimensional flow field is given by

$$\varphi = A\cos(\Omega t)\left(2x^2 - y^2 - z^2\right),$$

where A and Ω are known constants.

1. Determine the velocity field, verifying that $\nabla \wedge \vec{v} = 0$ and $\nabla \cdot \vec{v} = 0$.
2. Compute the streamlines, trajectories and particle paths.
3. Obtain the fluid surface that at the initial instant is a sphere of radius R centered at $(x, y, z) = (-R/2, 0, 0)$.

Source: K8.pdf (page 1). Notes.pdf, Chapter 2: potential Eq. (2.34); expansion rate Eqs. (2.37)-(2.38); trajectories and path lines Eqs. (2.12)-(2.15); fluid surfaces Eqs. (2.18)-(2.20); stream lines Eq. (2.21).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Three-dimensional, unsteady, potential flow, $\mathbf{v} = \nabla\varphi$, in all of space. The only time dependence is the common factor $\cos\Omega t$.
- Degrees of freedom: three coordinates (x, y, z); the field is linear in them.
- Units: φ is in m^2/s and multiplies a length squared, so A has units s^{-1}.
- At the instants when $\cos\Omega t = 0$ the velocity vanishes everywhere; streamlines are defined only at the other instants.

Symbol	Meaning	SI unit
A	amplitude of the strain rate	s^{-1}
Ω	angular frequency	s^{-1}
R	radius of the initial sphere	m
(x_i, y_i, z_i)	initial position of a particle	m
$s(t) = (A/\Omega)\sin\Omega t$	dimensionless time function, used below	1

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, origin at the centre of the field.
- Material (Lagrangian) variables: the initial position (x_i, y_i, z_i) of a particle. The trajectories give the map $(x_i, y_i, z_i) \mapsto (x, y, z)$ at time t, and its inverse is used in part 3.
- Parametrisation of the initial sphere (Notes, footnote to Eq. 2.19): $x_i = -R/2 + R\sin\alpha\cos\beta$, $y_i = R\sin\alpha\sin\beta$, $z_i = R\cos\alpha$, with $0 \le \alpha \le \pi$ and $0 \le \beta < 2\pi$. This is a change of variable from the two surface labels (α, β) to the three coordinates, eliminated at the end.

PHASE 3

Step-by-Step Derivation

3.1 Velocity field, curl and divergence

Why this tool: $\mathbf{v} = \nabla\varphi$ (Eq. 2.34). Let $c \equiv \cos\Omega t$.

$$v_x = \frac{\partial\varphi}{\partial x} = 4Acx, \qquad v_y = \frac{\partial\varphi}{\partial y} = -2Acy, \qquad v_z = \frac{\partial\varphi}{\partial z} = -2Acz,$$

$$\boxed{\mathbf{v} = Ac\,(4x,\ -2y,\ -2z).}$$

Curl, component by component:

$$(\nabla \wedge \mathbf{v})_x = \partial_y v_z - \partial_z v_y = 0 - 0, \quad (\nabla \wedge \mathbf{v})_y = \partial_z v_x - \partial_x v_z = 0 - 0, \quad (\nabla \wedge \mathbf{v})_z = \partial_x v_y - \partial_y v_x = 0 - 0,$$

so $\nabla \wedge \mathbf{v} = 0$ (as for any gradient). Divergence:

$$\nabla \cdot \mathbf{v} = \partial_x(4Acx) + \partial_y(-2Acy) + \partial_z(-2Acz) = Ac\,(4 - 2 - 2) = 0.$$

The potential satisfies $\nabla^2\varphi = Ac\,(4 - 2 - 2) = 0$: it is harmonic, the flow is irrotational and incompressible for all t. The extension along x is balanced by the compression along y and z.

3.2 Streamlines, trajectories and path lines

Why this tool: streamlines come from the tangency condition at a frozen instant (Eq. 2.21), trajectories from the initial-value problem $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12), and path lines from eliminating t (Eq. 2.15).

Streamlines (at an instant with $c \neq 0$ the common factor Ac cancels):

$$\frac{dx}{4x} = \frac{dy}{-2y} = \frac{dz}{-2z}.$$

From the last equality, $dy/y = dz/z \Rightarrow \ln|y/y_0| = \ln|z/z_0| \Rightarrow y/z = y_0/z_0$. From the first equality, $dx/x = -2\,dy/y \Rightarrow \ln|x/x_0| = -2\ln|y/y_0| \Rightarrow$

$$\boxed{x\,y^2 = x_0\,y_0^2, \qquad \frac{y}{z} = \frac{y_0}{z_0},}$$

the intersection of two surfaces (Eq. 2.15 type), independent of t.

Trajectories, with $\mathbf{x}(0) = (x_i, y_i, z_i)$. Each component decouples. Using Barrow's rule and $\int_0^t \cos\Omega t'\,dt' = \sin(\Omega t)/\Omega$:

$$\frac{dx}{dt} = 4A\cos(\Omega t)\,x \;\Rightarrow\; \int_{x_i}^{x}\frac{dx'}{x'} = 4A\int_0^t \cos\Omega t'\,dt' \;\Rightarrow\; \ln\frac{x}{x_i} = \frac{4A}{\Omega}\sin\Omega t,$$

$$\frac{dy}{dt} = -2A\cos(\Omega t)\,y \;\Rightarrow\; \ln\frac{y}{y_i} = -\frac{2A}{\Omega}\sin\Omega t, \qquad \frac{dz}{dt} = -2A\cos(\Omega t)\,z \;\Rightarrow\; \ln\frac{z}{z_i} = -\frac{2A}{\Omega}\sin\Omega t.$$

With $s(t) = (A/\Omega)\sin\Omega t$:

$$\boxed{x = x_i\,e^{4s(t)}, \qquad y = y_i\,e^{-2s(t)}, \qquad z = z_i\,e^{-2s(t)}.}$$

Path lines: eliminate s (that is, t). From the last two, $y/z = y_i/z_i$. Raise $y/y_i = e^{-2s}$ to the power -2: $(y/y_i)^{-2} = e^{4s} = x/x_i$, hence $xy^2 = x_i y_i^2$:

$$\boxed{x\,y^2 = x_i\,y_i^2, \qquad \frac{y}{z} = \frac{y_i}{z_i}.}$$

The path lines coincide with the streamlines of the flow (the direction field is steady; only the magnitude oscillates). The particles move back and forth along these curves: $s(t)$ oscillates between $\pm A/\Omega$, so x/x_i oscillates between $e^{-4A/\Omega}$ and $e^{4A/\Omega}$, and each particle returns to its initial position whenever $\sin\Omega t = 0$.

3.3 Fluid surface that is initially a sphere of radius R centred at $(-R/2, 0, 0)$
Why this tool: a fluid surface is the image under the trajectories of the initial surface (Eqs. 2.18-2.20); the surface labels (α, β) are then eliminated to obtain $f(x, y, z, t) = 0$.

Initial surface: $(x_i + R/2)^2 + y_i^2 + z_i^2 = R^2$, parametrised as in Phase 2. Trajectories map it to $x = x_i e^{4s}$, $y = y_i e^{-2s}$, $z = z_i e^{-2s}$, i.e. in terms of the labels,

$$\frac{x}{-R/2 + R\sin\alpha\cos\beta} = e^{4s}, \qquad \frac{y}{R\sin\alpha\sin\beta} = e^{-2s}, \qquad \frac{z}{R\cos\alpha} = e^{-2s}.$$

Eliminate (α, β) by inverting the map, $x_i = x\,e^{-4s}$, $y_i = y\,e^{2s}$, $z_i = z\,e^{2s}$, and substituting in the equation of the initial sphere:

$$\boxed{\left(x\,e^{-4s(t)} + \frac{R}{2}\right)^2 + \left(y\,e^{2s(t)}\right)^2 + \left(z\,e^{2s(t)}\right)^2 = R^2, \qquad s(t) = \frac{A}{\Omega}\sin\Omega t.}$$

This is an ellipsoid of revolution about the x-axis: its centre is at $x_c = -(R/2)\,e^{4s}$, its semi-axis along x is $R\,e^{4s}$ and its semi-axes along y and z are $R\,e^{-2s}$. The enclosed volume is $\frac{4\pi}{3}\,Re^{4s}\,R^2e^{-4s} = \frac{4\pi}{3}R^3$, constant, as required for an incompressible flow ($\nabla\cdot\mathbf{v} = 0$). At every instant with $\sin\Omega t = 0$ the surface is again the initial sphere.

Reading of the official handwritten solution

The statement fixes the centre at $(-R/2, 0, 0)$. In K8.pdf (part 3), as legible in the scan, the parametrisation is written $x_i - R/2 = R \sin\alpha \cos\beta$ and the final boxed equation contains $\left(x e^{-4A\sin(\Omega t)/\Omega} - R/2\right)^2$, which corresponds to a sphere centred at $x = +R/2$. For the sphere of the statement the sign in front of $R/2$ must be $+$, as derived above. Numerical test: for $R = 1$ m, $A = 0.6$ s^{-1}, $\Omega = 1.9$ s^{-1}, $t = 1.1$ s, a particle starting on the sphere of the statement ($\alpha = 0.8$, $\beta = 1.1$) satisfies the equation with $+R/2$ to 10^{-16}, whereas the $-R/2$ form leaves a residual of 0.349. The remaining handwritten results (components, streamlines, trajectories, paths) agree with the above.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: a pulsating pure-strain (stagnation-type) flow: the material is stretched along x and compressed in the (y, z) plane during the half cycles where $\cos\Omega t > 0$, and the opposite sense when $\cos\Omega t < 0$. The net displacement over a full period is zero.
- Limits: $A \to 0$ gives $s \to 0$ and the surface stays the initial sphere; $\Omega \to 0$ with $c \to 1$ gives the steady flow $\mathbf{v} = A(4x, -2y, -2z)$, for which $s \to At$ and the sphere is deformed without bound; at $\sin\Omega t = 0$ every trajectory returns to its starting point.
- Consistency: path lines equal streamlines for this flow, since the unit vector $\mathbf{v}/|\mathbf{v}|$ is time-independent; the volume of the fluid surface is conserved ($\nabla \cdot \mathbf{v} = 0$).
- Dimensions (SI): $[v] = [Ax] = \mathrm{s}^{-1}\mathrm{m} = \mathrm{m/s}$; $[s] = [A/\Omega] = 1$, so $e^{\pm ks}$ are dimensionless; the factors e^{4s}, e^{-2s} multiply lengths; in the ellipsoid equation each term has units m^2 ($x\,e^{-4s} + R/2$ is a length).
- Numerical cross-check: RK4 integration of the particle ODE with $A = 0.6$ s^{-1}, $\Omega = 1.9$ s^{-1} from $(0.5, 0.7, -0.4)$ m up to $t = 1.1$ s gives $(1.497122, 0.404533, -0.231162)$ m, the same as $x_i e^{4s}$, $y_i e^{-2s}$, $z_i e^{-2s}$ (agreement better than 10^{-11} m). Symbolic differentiation gives $\nabla \wedge \mathbf{v} = 0$ and $\nabla \cdot \mathbf{v} = 0$.